

Final Projects

1a: Integrals on $(-\infty, \infty)$ via standard contours

Evaluate the following using contour integration:

$$\int_{-\infty}^{\infty} \frac{\cos(3x)}{((x-1)^2 + 1)((x+2)^2 + 4)} dx$$

$$\int_{-\infty}^{\infty} \frac{\sin(5x)}{((x+1)^2 + 9)((x-3)^2 + 1)} dx$$

$$\int_{-\infty}^{\infty} \frac{\cos(2x)}{((x-2)^2 + 4)((x+1)^2 + 16)} dx$$

$$\int_{-\infty}^{\infty} \frac{\sin(7x)}{((x-1)^2 + 25)((x+2)^2 + 1)} dx$$

$$\int_{-\infty}^{\infty} \frac{\cos(4x)}{((x+3)^2 + 9)((x-2)^2 + 16)} dx$$

Notes:

Poles at $\alpha \pm ia$, $\beta \pm ib$

No poles on real or imaginary axes

Use e^{inx} and residues in the upper half-plane

1b: Integrals on $(0, \infty)$ via standard contours.

Evaluate the following using a keyhole contour:

$$\int_0^{\infty} \frac{x^{1/2}}{(x+1)((x-1)^2+4)} dx$$

$$\int_0^{\infty} \frac{x^{2/3}}{(x+2)((x+1)^2+1)} dx$$

$$\int_0^{\infty} \frac{x^{1/3}}{(x+3)((x-2)^2+9)} dx$$

$$\int_0^{\infty} \frac{x^{3/4}}{(x+1)((x+2)^2+4)} dx$$

$$\int_0^{\infty} \frac{x^{1/4}}{(x+2)((x-1)^2+16)} dx$$

Notes:

Branch cut on $(0, \infty)$

One pole at $x = -a$

Two poles at $\alpha \pm ib$

2: Conformal Mapping and Complex Potential Project

- a) Find a conformal map from the exterior of the unit circle to the exterior of a shape
- b) Construct several incompressible and irrotational flows in the circle domain
- c) Map these flows to the physical domain and illustrate them